

Supplementary Tables for Submission 991

Full supporting material (Set B)

This document supports our OpenReview response; the response is self-contained and this material is supporting only. All numbers use fresh rebuttal-period seeds and may differ from the submitted paper by up to about 0.04 on coverage and width, within seed variation. “Paper” denotes the submitted internal-split protocol (values quoted from the paper); “Strict” denotes the external 40/10/50 split.

Table B1. All fitted-model methods under the strict external split (R1)

R2CCP, CQR, and CHR under the submitted protocol and the strict external 40/10/50 split (3 judges, 10 seeds per strict-split cell; mean $\pm$ s.d.). Δ columns are paired t -test p -values on matched rebuttal-period runs. R2CCP coverage holds at 0.90 under both protocols; CQR over-covers throughout (weak quantile-regression features); CHR recovers nominal coverage under the strict split once the finite-sample $(1 + 1/n)(1 - \alpha)$ correction is included (standard split-CHR, Sesia & Romano, 2021).

Method	Judge	Paper		Strict 40/10/50		Paired p	
		cov	w	cov	w	Δcov	Δw
R2CCP	LLaVA-Critic	0.900	3.05	0.894 ± 0.015	3.041 ± 0.095	0.772	0.093
R2CCP	Phi-4	0.891	3.13	0.890 ± 0.012	3.145 ± 0.076	0.377	0.908
R2CCP	Gemini	0.898	2.85	0.897 ± 0.012	2.887 ± 0.067	0.566	0.015
CQR	LLaVA-Critic	0.996	3.95	0.970 ± 0.011	3.60	0.002	0.002
CQR	Phi-4	0.997	3.98	0.993 ± 0.009	3.91	0.641	0.469
CQR	Gemini	0.996	3.95	0.977 ± 0.012	3.65	0.002	0.002
CHR	LLaVA-Critic	0.880	2.97	0.900 ± 0.014	3.29	0.002	0.002
CHR	Phi-4	0.896	3.16	0.902 ± 0.015	3.33	0.004	0.002
CHR	Gemini	0.880	2.80	0.906 ± 0.012	3.20	0.002	0.002

Paper figures are quoted from Table 1 (cross-judge R2CCP and CHR rows), Table 2 (LLaVA-Critic CQR, 0.996/3.95), and Table 9 (cross-judge CQR, 0.996/0.997/0.996).

Boundary-adjustment implementation (R2)

For completeness, the boundary-adjustment routine applies the expansion form $[l, u] \mapsto [[l], [u]]$, not the shrinkage form printed in Sec. 3.3. This is consistent with the expansion branch of the

cited Theorem 1 (Sheng et al., 2025) and with the reported increase in coverage and width after adjustment (LLaVA-Critic R2CCP: 0.900/3.05 $\rightarrow$ 0.981/3.60).

```
def boundary_adjust_expand(lows, highs):  
    # Expand: floor lower, ceil upper  
    return np.clip(np.floor(lows), 1, 5), np.clip(np.ceil(highs), 1, 5)
```